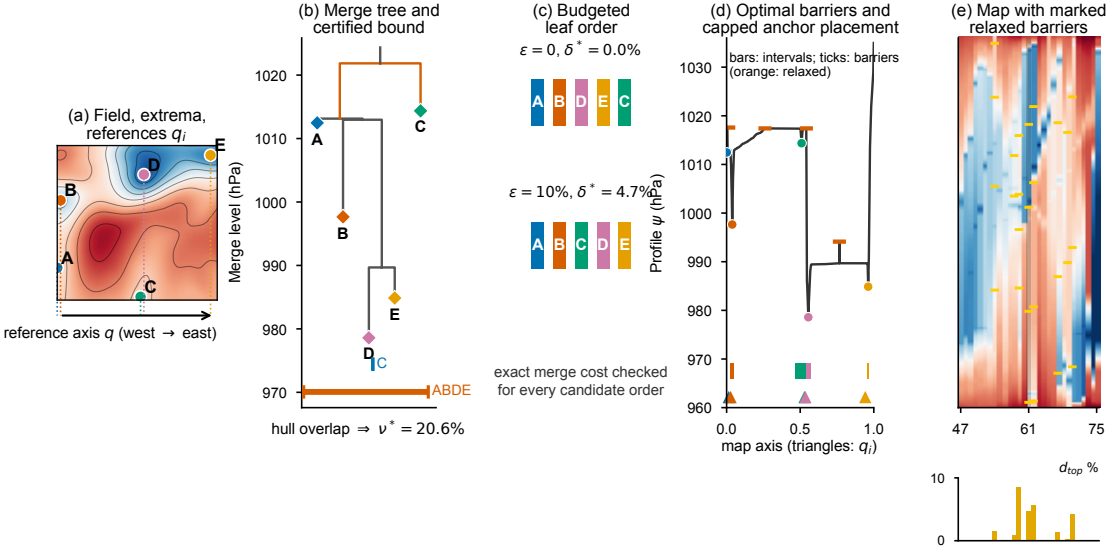

(a) Field, extrema, references q_i
A
B
C
D
E
reference axis q (west → east)
(b) Merge tree and certified bound
Merge level (hPa)
1020
1010
1000
990
980
970
A
B
C
D
E
ABDE
hull overlap ⇒ $\nu^* = 20.6\%$
(c) Budgeted leaf order
$\varepsilon = 0$, $\delta^* = 0.0\%$
A B D E C
$\varepsilon = 10\%$, $\delta^* = 4.7\%$
A B C D E
exact merge cost checked for every candidate order
(d) Optimal barriers and capped anchor placement
bars: intervals; ticks: barriers (orange: relaxed)
Profile ψ (hPa)
1030
1020
1010
1000
990
980
970
960
0.0
0.5
1.0
map axis (triangles: q_i)
(e) Map with marked relaxed barriers
47
61
75
d_{top} %
10
0